

Gurpreet Singh

Lead Data Engineer — Cloud Data Platforms & Real-Time Analytics

Gurugram, Haryana, India | +91 76784 57429 | gurpreet040888@gmail.com | 15+ years experience

15+

Years in data engineering

40%

Query latency cut after Snowflake migration

70%

Faster query turnaround via NLQ

10+

Engineers led & mentored

CAREER TIMELINE · 2010 – PRESENT

	AON Hewitt	Amdocs	Nagarro	Expedia Group	Amex GBT - Current
Nucleus Software Intern · 5 mo	Accenture 14 mo	AON Hewitt 27 mo	Amdocs 25 mo	Nagarro 24 mo	
Amdocs 25 mo	Nagarro 24 mo	Expedia Group (Egencia) 55 mo	American Express GBT 44 mo · current		

PROFESSIONAL SUMMARY

Lead Data Engineer with 15+ years of experience architecting and implementing scalable cloud-based data platforms and near real-time analytics solutions **for product companies**. Proven expertise across **AWS** (Kinesis, Lambda, Glue, EMR), **Snowflake**, **Redshift** and enterprise ETL frameworks to deliver high-performance data pipelines and business intelligence systems. Strong communicator with a track record of leading cross-functional engineering teams and delivering enterprise-grade data solutions that improve efficiency and decision-making.

CORE SKILLS

Cloud & Big Data	AWS	Kinesis	Lambda	Glue	EMR / EMR Serverless	EC2	S3	DynamoDB	Airflow	SQS / SNS
Data Platforms	Snowflake	Redshift	Athena	Teradata	SQL Server	Oracle				
ETL & Orchestration	Informatica	SSIS	Stonebranch	Jenkins	CI/CD					
Programming	Python	PySpark	Unix Shell	PL/SQL	Excel	VBA				
Governance & Observability	Alation	ELK Stack	DataDog							
BI & Reporting	Sisense	Tableau	Excel							

CERTIFICATIONS

AWS Certified Solutions Architect	2020	Python / Data Science Training — Edureka	2019	Big Data and Hadoop — Edureka	2015
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EDUCATION

B.Tech, Computer Science & Engineering

Guru Tegh Bahadur Institute of Technology (GGSIPU)
2010 · 77.34%

CBSE Senior Secondary (12th)

Army Public School, Dhaula Kuan
82.00%

LEADERSHIP & INNOVATION HIGHLIGHTS

- Extensive experience in product-based companies, driving innovation and customer-focused data solutions.
- Lead and mentor a team of **10+** data engineers, driving technical excellence and career growth.
- Collaborate closely with Product Managers and Directors for 180-day planning, estimation, and delivery of enterprise data initiatives.
- Define and manage cross-team dependencies, ensuring alignment with organizational priorities and timely execution.
- Partner with leadership to set strategic priorities and roadmap for data platform evolution.
- Spearhead initiatives to build AI competencies within the data engineering team.
- Lead development of AI-powered dashboards to enhance analytics and decision-making capabilities.
- Drive adoption of machine learning and predictive analytics in BI reporting for advanced insights.

PROFESSIONAL EXPERIENCE

Lead Data Engineer

Dec 2022 – Present

American Express Global Business Travel (Egencia) · Gurugram, India

- Lead design and development of data warehouses for B2B corporate travel analytics within the Enterprise Data Platform team.
- Delivered multiple projects independently and as team lead; set up agile processes and project estimation frameworks.
- Hands-on across AWS services: Redshift, Lambda, EMR clusters, S3, EC2, IAM, CloudWatch, KMS, VPC, DynamoDB, Aurora.
- Designed and developed ETL solutions to ingest and consume data from multiple sources into a Snowflake-based DWH.
- Migrated multi-TB volumes of data to Snowflake, cutting query latency by **40%**.
- Enabled Natural Language Querying for customer data, reducing query turnaround time by **70%**.
- Developed ML-driven, recommendation-based dashboards delivering predictive performance metrics for decision-making.
- Key contributor to a new cloud DWH consuming Protobuf transactional messages from Kinesis, streamed via PySpark/EMR/Python into Snowflake — publishing data near real-time for downstream analytics.
- Built and tuned DWH on Snowflake & Redshift; wrote Redshift queries with sort/distribution keys for transformation performance.
- Built visualizations and reports in Sisense on a multi-node cluster to support client reporting (QBR, Travel Summary).
- Built Kibana dashboards on the ELK Stack for data reconciliation.

Senior Data Engineer

May 2018 – Dec 2022

Expedia Group (Egencia) · Gurugram, India

- Owned the end-to-end lifecycle: requirement analysis, solution design, delivery, and production operations.
- Established Data Governance within the organization and onboarded Alation to streamline BI metrics across the BI Group.
- Implemented streaming ingestion from Kinesis to Snowflake via PySpark/EMR, enabling near real-time publishing.
- Built Python/PySpark utilities for RDBMS-to-Snowflake/Redshift ingestion and transformation.
- Created data models and leveraged the cloud platform for data processing.
- Authored Redshift SQL (distribution/sort keys) for performant transformations and marts.
- Created Sisense dashboards across a multi-node cluster for client reporting (QBR, Travel Summary).
- Configured Kibana dashboards for data reconciliation; contributed to agile processes and estimation.
- Promoted within **18 months** of joining.

Technical Lead

May 2016 – May 2018

Nagarro · Gurugram, India

- ETL development for Utilities and Finance domains (Ecova, McKinsey Financial Analytics).
- Led a team of **7+** developers, working closely with the Product Manager and business analysts.
- Implemented Informatica-based pipelines; managed admin/tuning and migration from SSIS to Informatica.
- Designed and developed Oracle PL/SQL ETL processes.
- Built reports analyzing budgeted vs. actual expense per engagement (WIP/iWIP, Billing Invoice, FP&A).

Senior Subject Matter Expert

Apr 2014 – May 2016

Amdocs · Gurugram, India

- Over **1 year** onsite handling the Informatica environment as Admin and developer for Globe Telecom (Philippines) and Optus (Australia).
- Designed logical and physical data models (Erwin) and Informatica logic for CDC and staging layers.
- Utilized Teradata utilities (FEXP, FLOAD, MLOAD, TPT); automated exports/loads across environments.
- Built Unix shell scripts to orchestrate Informatica workflows and DDL/DML automation.
- Managed Dev/Test/Prod environments; configured Hadoop user matrix and HDFS/Hive/HBase privileges.

Senior Software Engineer → Software Engineer

Jan 2012 – Apr 2014

AON Hewitt · Gurugram, India

- Developed Informatica workflows to load client data for Total Rewards (Payroll, Incentives, Stocks).
- Authored complex mappings generating XML payloads pushed to database/LDAP.
- Built Unix wrapper scripts and Control-M schedules; created utilities (XLS2CSV, Perf Analyzer).
- Implemented a web analytics warehouse (Activity Reporting): staging, ODS with history, dimensions, facts.
- Delivered the HMCGE data warehouse; Oozie jobs; Pig/HiveQL implementations.
- Awards: Premier Team Gold, Gold Award for Execution; promoted within 18 months.

Other Work Experience

Associate Software Engineer, Accenture — Nov 2010 – Jan 2012 · Lloyds Banking Group (Banking), Sigma Aldrich (Utilities)

Project Intern, Nucleus Software — Jun 2010 – Nov 2010 · Cash@Will (Core Java Developer)

Gurpreet Singh · Lead Data Engineer · Gurugram, India